

an LM7812 voltage regulator (IC1), a CD4060 CMOS counter IC (IC2), and a few discrete components.

The circuit operates from the 230V AC mains and provides isolated, regulated DC power along with multiple selectable delay outputs for driving a relay. The 230V AC mains input is stepped down to 12-0-12V AC using a centre-tapped transformer (X1). Diodes D1 and D2 form a full-wave rectifier, converting the AC into pulsating DC, which is filtered by electrolytic capacitor C1 to obtain a smooth DC output. This unregulated DC is fed to LM7812 voltage regulator (IC1), which provides a stable 12V DC output for the relay and driver stages. Capacitors C1 and C2 are used at the regulator's input and output for stability and ripple suppression. LED1 indicates the presence of the regulated supply.

The regulated 12V supply powers the CD4060 (IC2). Its internal oscillator frequency is determined by resistors R2 and R3 and capacitor C3. Once powered, the oscillator starts automatically and the counter begins dividing the clock signal to generate progressively increasing time delays at outputs Q4-Q14.

With the CD4060 configured with $R2 = 2M\Omega$ (pin 10), $R3 = 120k\Omega$ (pin 11), and $C3 = 4.7\mu F$ (pin 9), as shown in Fig. 2, the timing outputs appear sequentially. After reset, Q4 becomes the first output to go high at about 24 seconds, followed by Q5 at 48 seconds, Q6 at 96 seconds, Q7 at 192 seconds, Q8 at 6.4 minutes, Q9 at 12.8 minutes, Q10 at 25.6 minutes, Q11 at 51.2 minutes, Q12 at 1.7 hours, Q13 at 3.4 hours, and Q14 at approximately 6.8 hours.

The table shows the rotary switch (S3) position and the corresponding CD4060 output delay. These timing outputs are connected to the common pin 13 of the multi-position rotary switch S3, which selects one output at a time.

The working principle of the device is straightforward. When start switch S1 is pressed momentarily, two actions occur simultaneously. First, the appliance connected across CON2 is switched on. Second, the timer IC CD4060 begins counting from the reset (zero) state.

After the selected delay time elapses, the corresponding output of IC2 goes high. For example, when the rotary switch S3 is set to position 2, the Q5 output becomes high after the preset delay (here 48 seconds). This high signal is routed through the common terminal (pin 13) of switch S3 to the base of transistor T1 through resistor R5. As a result, transistor T1 conducts, which in turn switches off transistor T2.

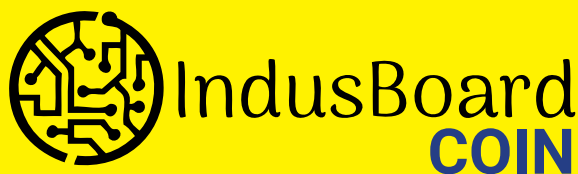
When transistor T2 stops conducting, relay RL1 becomes de-energised, causing the appliance (load) connected to its contacts to switch off automatically after the set delay (here 48 seconds). The stop switch



An "AWESOME" Development Board from India?

- **Super Small:** 3 centimeters
- **Super Smart:** 3 processors
(Xtensa® Microprocessor + 2 ULP RISC-V Coprocessor)
- **Super Sensitive:** 5 Sensors
- **Super Connected:** Wifi & 10+ Interfaces
- **Super Flexible:** 30 GPIOs
- **Super Easy:** FPC + Dedicated Battery Connector

Introducing the



Priced Shockingly at ₹2,359 (Inclusive of GST)



SPECIAL INTRO PRICE

₹2,199 (Inclusive of GST)

KNOW MORE / BUY NOW: indus.electronicsforu.com

Developed by the Electronics For You team, for you.

DEALERS: • Om Technology Centre, Bengaluru (Mob: +919845400864)
• Bharathi Electronics: Coimbatore (Mob: +919952228618)

